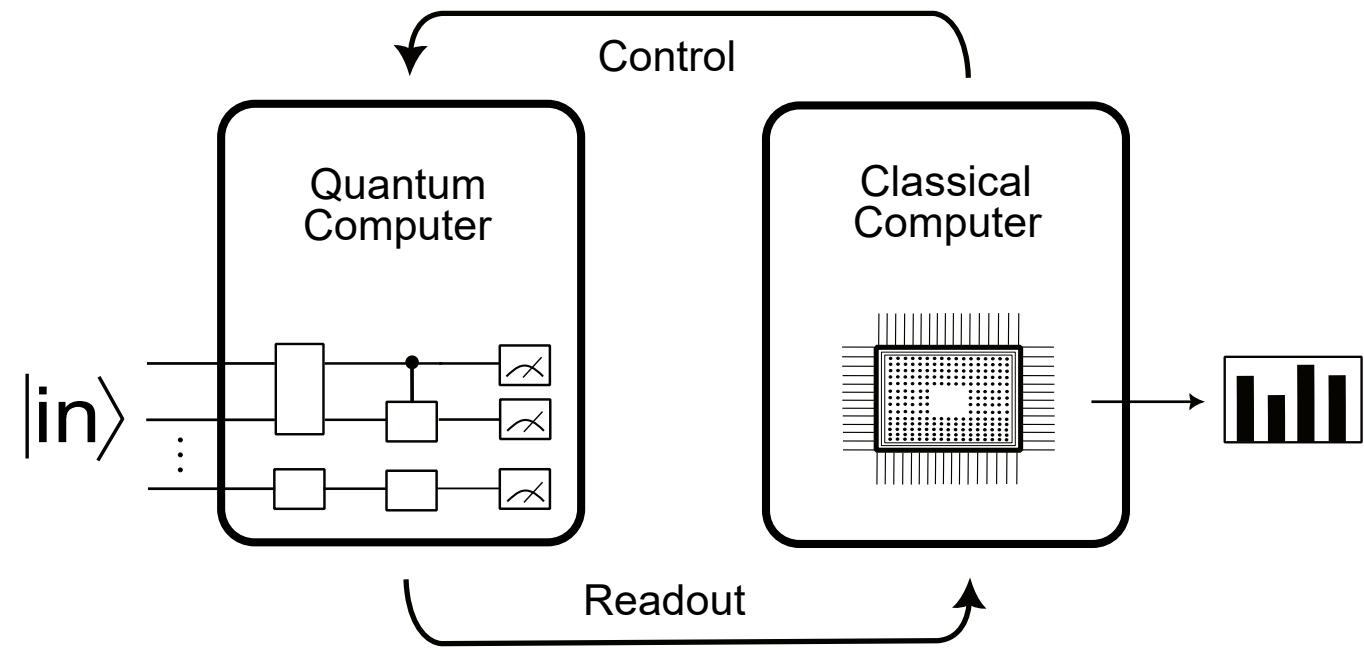


## Motivation: Quantum and Classical Together

**Open** quantum systems interact with their environment, leading to **non-unitary** evolution. Quantum hardware executes only **unitary** gates  $\Rightarrow$  embed non-unitary matrices into unitary ones



Hybrid loop: the CPU builds and controls the circuit, the QPU carries the exponentially large state, the measurement is read back.

- **Scaling:**  $2^n$  amplitudes for  $n$  qubits
- **Task:** embed a non-unitary matrix into a larger unitary one
- **Tool:** Kraus operators + SVD + dilation
- **Test system:** the FMO light-harvesting complex

## Qubits and the Bloch Sphere

A qubit may be in a **superposition** of the basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Normalised,  $|\alpha|^2 + |\beta|^2 = 1$ , and  $\text{Pr}(0) = |\alpha|^2$ .

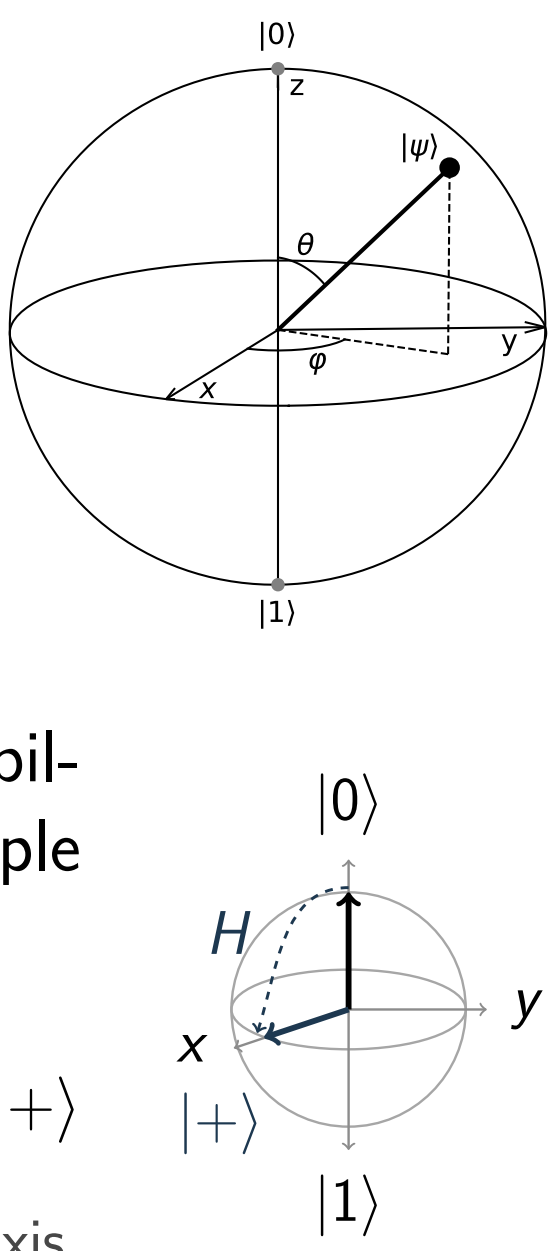
With  $\alpha = \cos \frac{\theta}{2}$  and  $\beta = e^{i\varphi} \sin \frac{\theta}{2}$ , the normalisation becomes two angles:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle$$

**A gate is a rotation of a state.** Probability conservation demands  $U^\dagger U = \mathbb{1}$ . Example – the Hadamard gate applied to  $|0\rangle$ :

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \equiv |+\rangle$$

The arrow swings from the north pole onto the x axis.

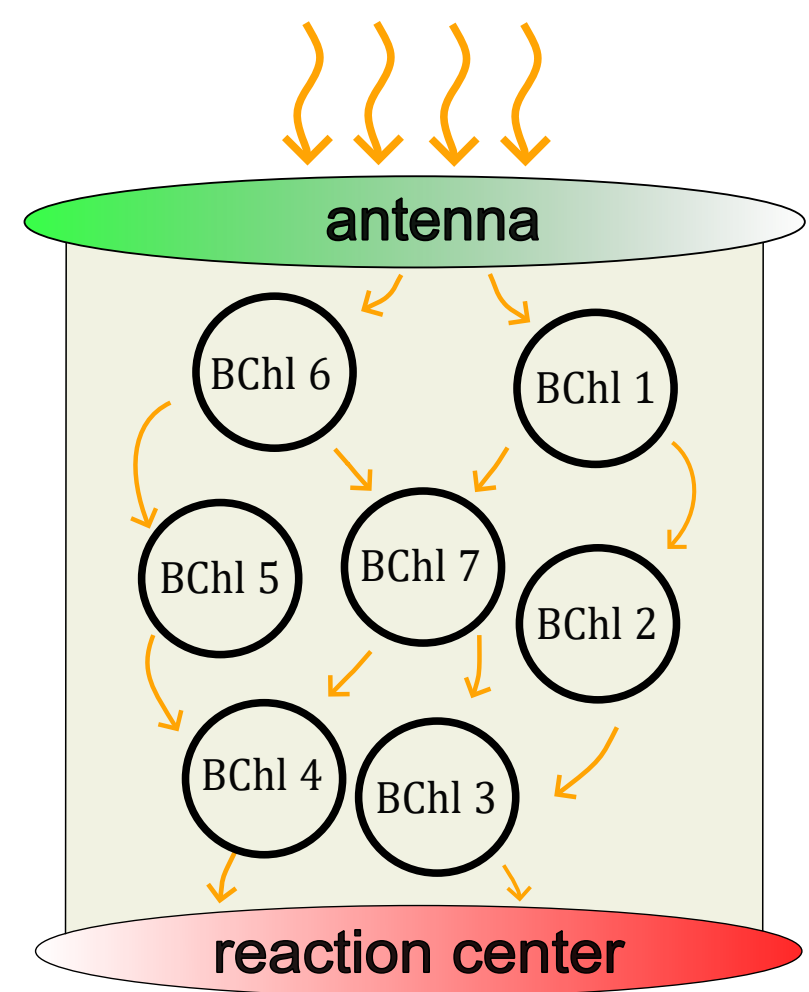


## The Fenna–Matthews–Olson Complex

Seven **bacteriochlorophyll** (BChl) molecules guide an excitation from the antenna to the reaction center [1].

- The excitation **delocalises** over all sites (dipole–dipole interaction)
- Each BChl is a **two-level system** (HOMO/LUMO). With at most one excitation in the complex this gives the site basis:

$$|i\rangle = |\phi_1^g\rangle \cdots |\phi_i^e\rangle \cdots |\phi_N^g\rangle$$



Single monomer of the FMO complex; the arrows show the excitation transfer.

## The Frenkel Exciton Model

Following Kreisbeck [2] the Hamiltonian splits into three parts:

$$\hat{H} = \hat{H}_S + \hat{H}_B + \hat{H}_{SB}$$

**System** – site energies and dipole–dipole coupling:

$$\hat{H}_S = \sum_m \epsilon_m^e |m\rangle \langle m| + \sum_{m \neq n} J_{mn} |m\rangle \langle n|$$

$J_{mn}$ : excitation hops  $m \rightarrow n \Rightarrow$  delocalisation. No electrons are exchanged.

**Bath** – nuclear vibrations as harmonic normal modes:

$$\hat{H}_B = \sum_{\xi} \left( \frac{p_{\xi}^2}{2m_{\xi}} + \frac{1}{2} m_{\xi} \omega_{\xi}^2 q_{\xi}^2 \right) = \sum_{\xi} \hbar \omega_{\xi} \left( b_{\xi}^\dagger b_{\xi} + \frac{1}{2} \right)$$

$b_{\xi}^\dagger$  creates a phonon  $\hbar \omega_{\xi}$  in mode  $q_{\xi}$ .

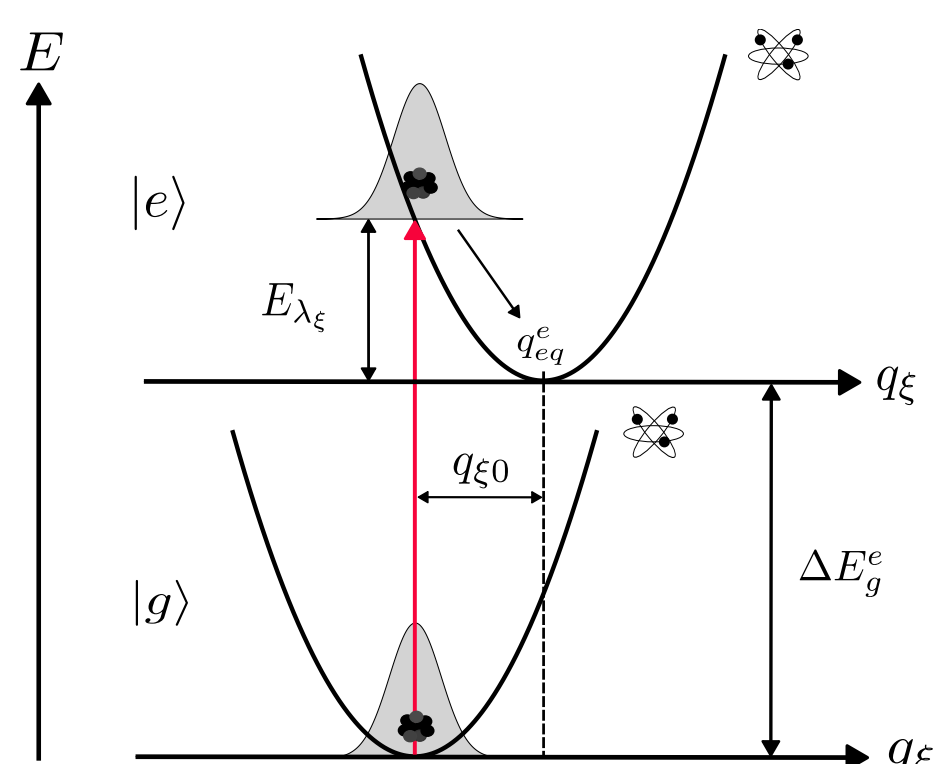
**Coupling** – exciton–phonon coupling equilibrium shift:

$$\hat{H}_{SB} = \sum_m \sum_{\xi} \hbar \omega_{\xi m} d_{\xi m} (b_{\xi m}^\dagger + b_{\xi m}) \otimes |m\rangle \langle m|$$

- Nuclear dynamics follow potential energy surfaces.

- In the excited state the equilibrium is shifted by  $q_{\xi 0}$ .

- Relaxation into the new minimum dissipates the **reorganisation energy**.



Red arrow: Vertical Franck–Condon transition: nuclei remain fixed during excitation ( $\tau_{el} \ll \tau_{nuc}$ ), since electronic dynamics is much faster than nuclear motion.

## Markovian Dynamics: Lindblad $\rightarrow$ Kraus

**Lindblad equation** – the most general Markovian master equation. Uses Born, Markov and secular approximation:

$$\frac{d}{dt} \rho_S = \underbrace{-\frac{i}{\hbar} [H, \rho_S]}_{\text{unitary}} + \underbrace{\sum_k \gamma_k \left( L_k \rho_S L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho_S\} \right)}_{\text{dissipator} \rightarrow \text{non-unitary}} := \mathcal{L} \rho_S$$

$\gamma_k$ : transition rate (Fermi's golden rule).  $L_k$ : Lindblad operator (jump operator).

1. **Vectorise** the Lindblad equation  $\Rightarrow$  obtain map  $\Lambda(t)$ :

$$\text{vec}(\rho_S(t)) = \Lambda(t) \text{vec}(\rho_S(0))$$

2. **Choi matrix** of the map  $\Lambda(t)$ , then eigendecomposition of  $C(t)$ :

$$C(t) = \sum_{i,j} (E_{ij} \otimes \mathbb{1}) \Lambda(t) (\mathbb{1} \otimes E_{ij}) = \sum_k \sigma_k u_k u_k^\dagger$$

3. **Kraus operators** from eigenvectors and eigenvalues of  $C(t)$ :

$$\text{vec}(M_k(t)) = \sqrt{\sigma_k(t)} u_k(t), \quad \rho_S(t) = \sum_k M_k(t) \rho_S(0) M_k^\dagger(t)$$

## Construction of the Circuit [4]

The  $M_k$  are **not unitary**. An SVD isolates the non-unitary part in the diagonal matrix  $\Sigma$ :

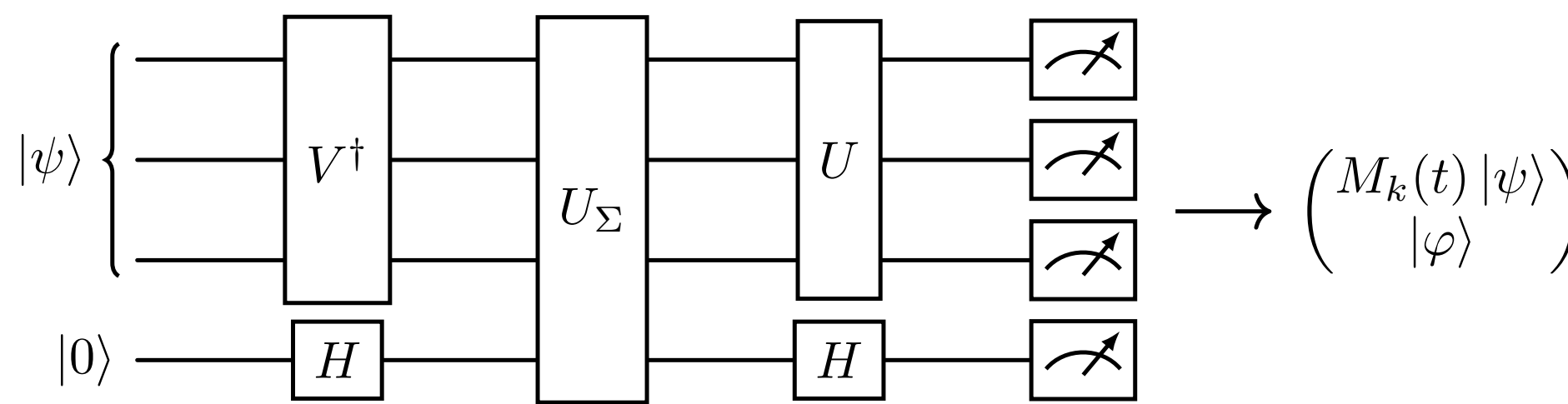
$$M_k(t) = U \Sigma V^\dagger$$

$U, V^\dagger$  are rotations and already unitary – only the rescaling  $\Sigma$  must be re-prepared.

**Dilation** into a unitary of twice the size adds one ancilla qubit:

$$U_\Sigma = \begin{pmatrix} \Sigma_+ & 0 \\ 0 & \Sigma_- \end{pmatrix}, \quad \Sigma_{\pm ij} = \sigma_{ij} \pm i \sqrt{1 - \sigma_{ij}^2}$$

$|\sigma_{ij}| \leq 1$  is guaranteed by complete positivity, so the root stays real.

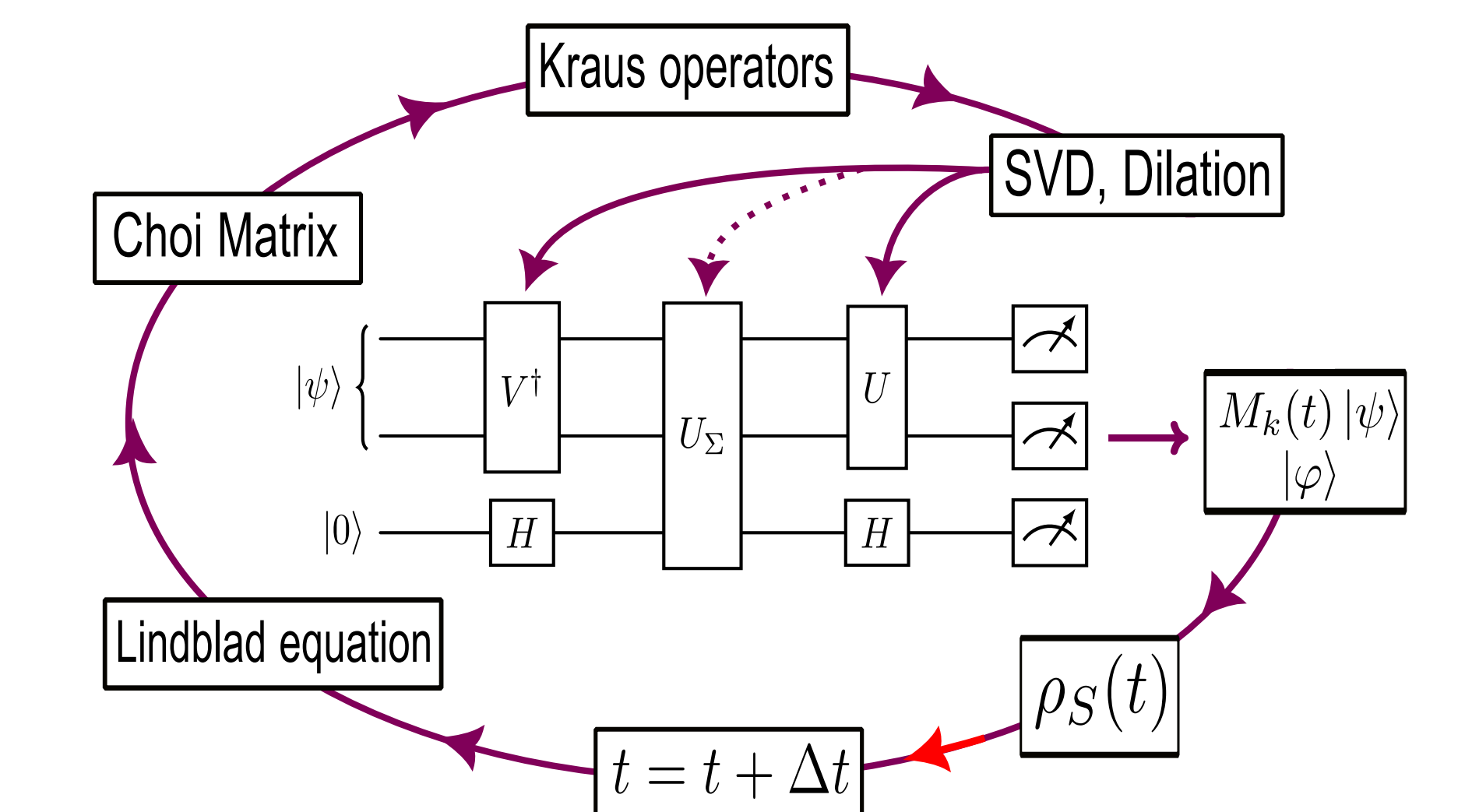


Seven-site FMO: three system qubits carry  $|\psi\rangle$ , the fourth is the ancilla, framed by two Hadamards.

Propagating the initial state  $|\psi\rangle \otimes |0\rangle$  through the circuit and using  $\Sigma_+ + \Sigma_- = 2\Sigma$ :

$$U H U_\Sigma V^\dagger H |\psi\rangle \otimes |0\rangle = \frac{1}{2} \begin{pmatrix} U(\Sigma_+ + \Sigma_-) V^\dagger |\psi\rangle \\ U(\Sigma_+ - \Sigma_-) V^\dagger |\psi\rangle \end{pmatrix} = \begin{pmatrix} M_k(t) |\psi\rangle \\ |\varphi\rangle \end{pmatrix}$$

**The upper half of the output is exactly  $M_k(t)|\psi\rangle$ ;  $|\varphi\rangle$  is a residue of the dilation and is discarded.**

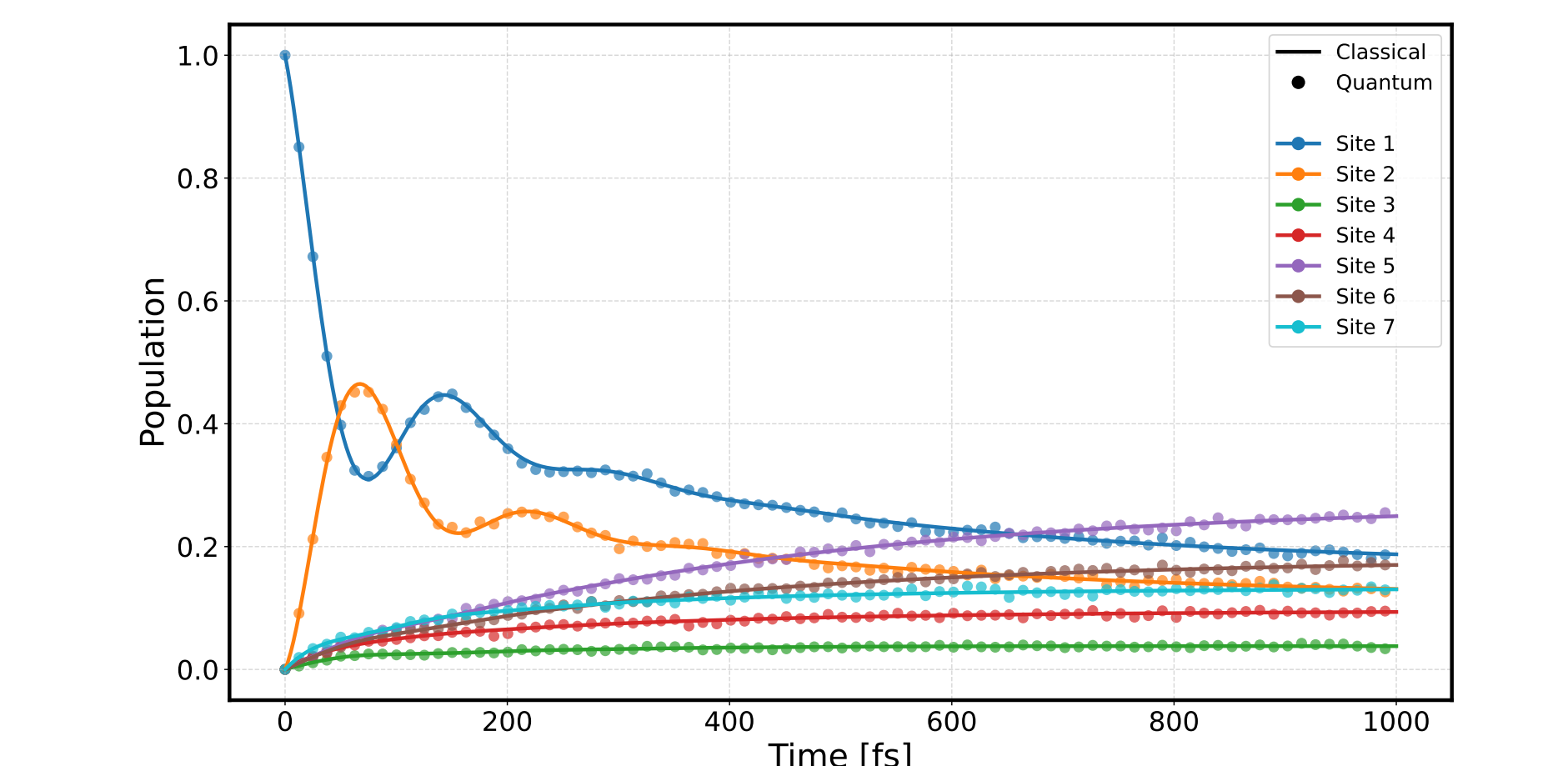


The loop: Lindblad  $\rightarrow$  Choi  $\rightarrow$  Kraus  $\rightarrow$  SVD and dilation  $\rightarrow$  circuit  $\rightarrow \rho_S(t)$ , then one step further in time.

**The enhanced algorithm.** Do not re-prepare the state classically but reuse the quantum state from the *previous* step with QST:

- the **same** circuit is applied again and again
- classical cost independent of  $t \Rightarrow$  work moves to QPU

## Result: Markovian Dynamics



Site populations of the seven-site FMO complex over 1000 fs. Lines: classical reference, dots: quantum simulation (on a classical computer) using Qiskit.

## Non-Markovian Dynamics: HEOM [3]

Unlike the Markovian case, the **hierarchical equations of motion** capture bath memory by absorbing bath modes into a coupled hierarchy of auxiliary density operators (ADOs)  $\hat{\sigma}_n$ :

$$\frac{d}{dt} \hat{\sigma}_n = -\underbrace{\frac{i}{\hbar} \mathcal{L}_S \hat{\sigma}_n}_{\text{own dynamics and decay}} + \underbrace{\sum_{j,k} (\phi_j \theta_{j,0} - \eta_{jk} \gamma_{jk}) \hat{\sigma}_n}_{\text{level above}} + \underbrace{\sum_{j,k} \phi_j \hat{\sigma}_{n_{j,k}}}_{\text{level below}} + \underbrace{\sum_{j,k} \eta_{jk} \theta_{j,k} \hat{\sigma}_{n_{j,k}}}_{\text{level below}}$$

$\phi_j, \theta_{j,k}$ : bath-coupling superoperators,  $\gamma_{jk}$ : decay rate of mode  $jk$ . Level 0 is  $\rho_S$  itself.

**Numerically exact** – no Born, Markov or secular approximation; the only error is the truncation depth  $\mathcal{N}$ :

$$\# \text{ ADOs} = \frac{(\mathcal{N} + NK)!}{\mathcal{N}! (NK)!}$$

$N$  sites,  $K$  bath exponentials – grows fast, HEOM is expensive.

## The Non-Markovian Propagator on a Circuit

Following Dan *et al.* [5]: HEOM yields a propagator  $G(t)$  whose  $m$ -th column holds the populations for an initial excitation on site  $m$ :

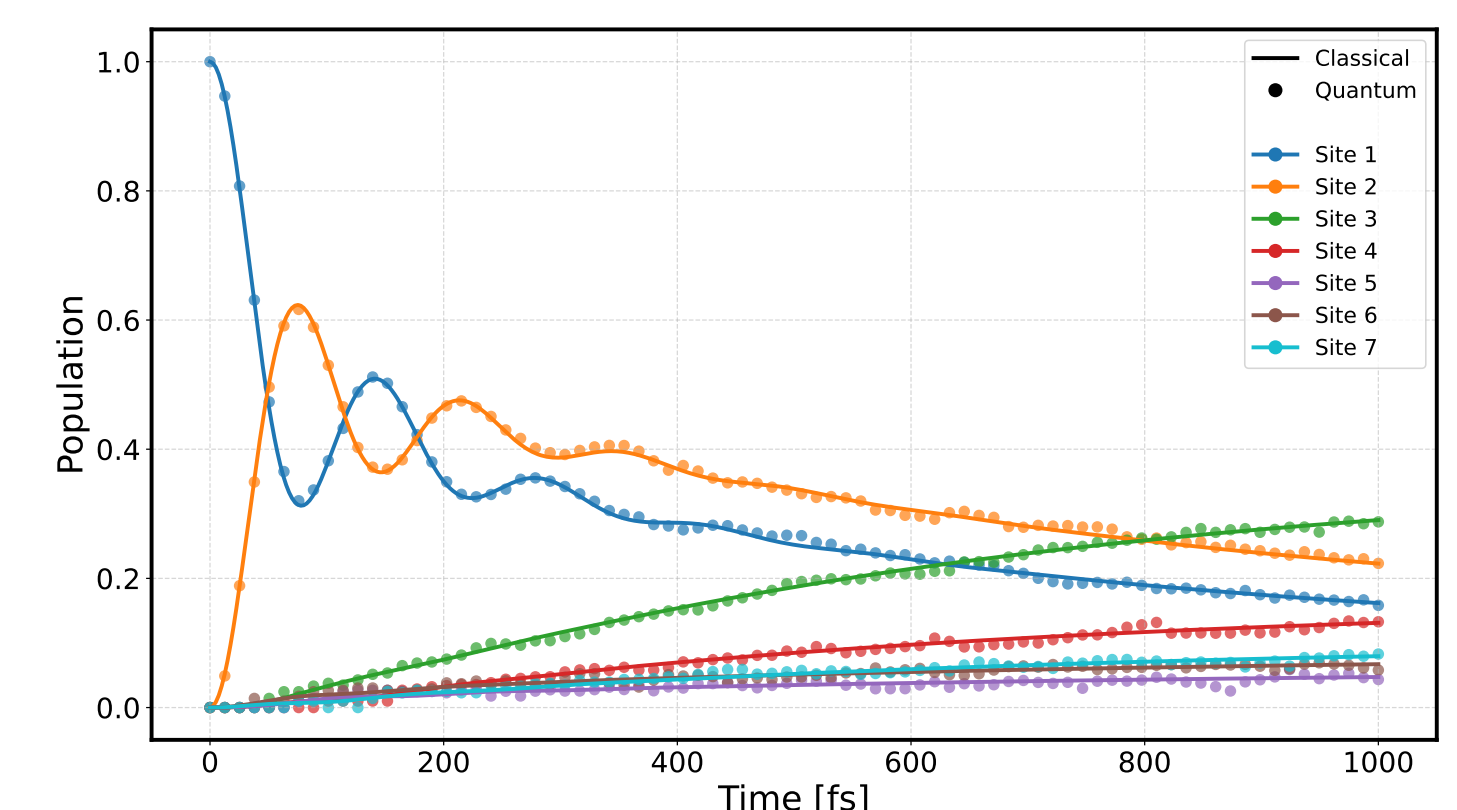
$$G(t) = \begin{pmatrix} \rho_{11}^1(t_0) & \rho_{11}^2(t_0) & \rho_{11}^3(t_0) & \rho_{11}^4(t_0) \\ \rho_{22}^1(t_0) & \rho_{22}^2(t_0) & \rho_{22}^3(t_0) & \rho_{22}^4(t_0) \\ \rho_{33}^1(t_0) & \rho_{33}^2(t_0) & \rho_{33}^3(t_0) & \rho_{33}^4(t_0) \\ \rho_{44}^1(t_0) & \rho_{44}^2(t_0) & \rho_{44}^3(t_0) & \rho_{44}^4(t_0) \end{pmatrix} \begin{pmatrix} \rho_{11}^1(t_1) & \rho_{11}^2(t_1) & \rho_{11}^3(t_1) & \rho_{11}^4(t_1) \\ \rho_{22}^1(t_1) & \rho_{22}^2(t_1) & \rho_{22}^3(t_1) & \rho_{22}^4(t_1) \\ \rho_{33}^1(t_1) & \rho_{33}^2(t_1) & \rho_{33}^3(t_1) & \rho_{33}^4(t_1) \\ \rho_{44}^1(t_1) & \rho_{44}^2(t_1) & \rho_{44}^3(t_1) & \rho_{44}^4(t_1) \end{pmatrix}, \dots$$

Four-site case. Superscript: site of the initial excitation, one matrix per time step.

$G(t)$  is non-unitary, so the same SVD and the same dilation  $U_\Sigma$  as in the Markovian case apply:

$$G(t) = U \Sigma V^\dagger, \quad |\psi(t)\rangle = G(t) |\psi(0)\rangle$$

This yields:



Non-Markovian seven-site FMO: the coherent oscillation between sites 1 and 2 survives longer.

## Continuous Grid on Real Hardware

**Approach.** Stack  $\rho_S$  and all ADOs into one super-vector  $\vec{\Sigma} = (\text{vec } \rho_S, \text{vec } \sigma_1, \dots)^\top$  and use the RHS of HEOM as a propagator:

$$\frac{d}{dt} \vec{\Sigma} = \mathcal{L}_{\text{HEOM}} \vec{\Sigma} \Rightarrow P = e^{\mathcal{L}_{\text{HEOM}} \Delta t}, \quad \vec{\Sigma}_n = P^n \vec{\Sigma}_0$$

**Obstacle.**  $\|P\|_2 = 32.39 \Rightarrow$  a dilation of  $P$  gives a small success probability  $\Rightarrow$  many shots on the QPU.

**Fix.** Use the Lyapunov equation to find  $W$  for a basis transformation:

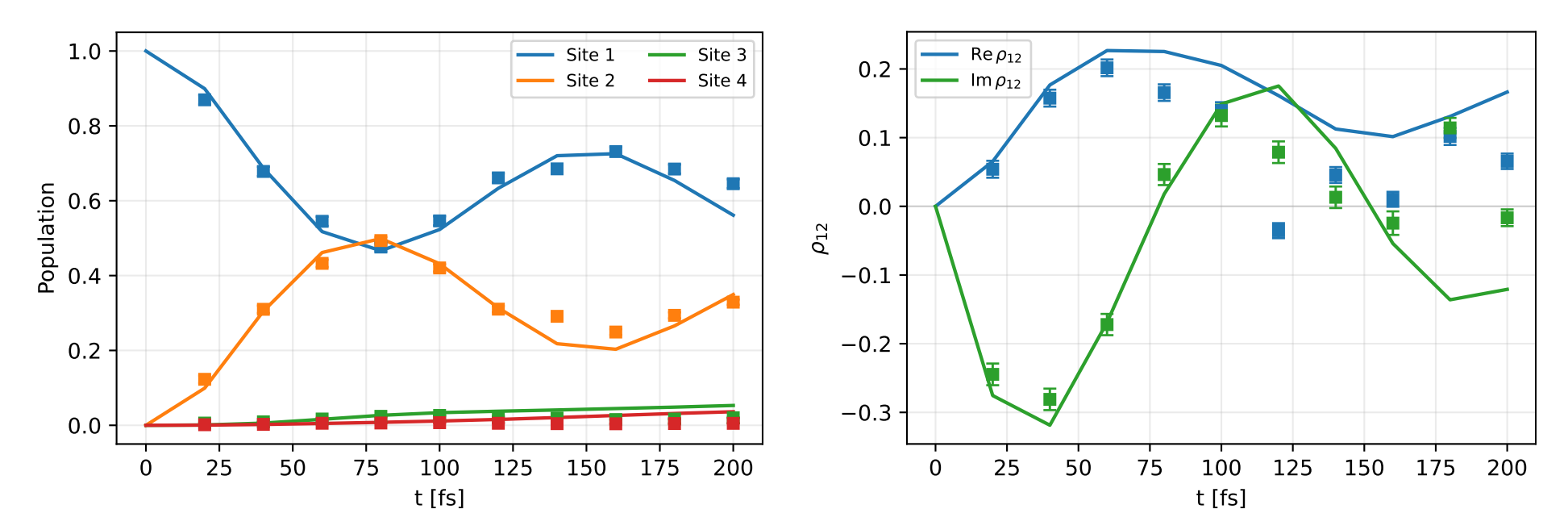
$$\tilde{P} = W^{1/2} P W^{-1/2}, \quad \|\tilde{P}\|_2 \leq e^{\delta \Delta t} \approx 1$$

**Compression.**  $P$  is too large for current quantum hardware  $\Rightarrow$  compress it to Krylov space,  $\tilde{H}_m = \tilde{Q}_m^\dagger \tilde{P} \tilde{Q}_m$ , and dilate that:

$$U = \begin{pmatrix} \tilde{H}_m/s & \sqrt{\mathbb{1} - \tilde{H}_m \tilde{H}_m^\dagger / s^2} \\ \sqrt{\mathbb{1} - \tilde{H}_m^\dagger \tilde{H}_m / s^2} & -(\tilde{H}_m/s)^\dagger \end{pmatrix}$$

$m = 4 \Rightarrow$  **3 qubits**, 31 CNOTs per step: **one single unitary**, re-applied  $n$  times.

**On the quantum processor.** The classical cost no longer grows with the number of steps since the same circuit is applied repeatedly:



Four-site FMO on ibm\_kingston, 70 circuits,  $\Delta t = 20$  fs. Left: site populations, right: real and imaginary part of the coherence  $\rho_{12}$ . Lines: classical reference, squares: hardware.

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- [1] J. Adolphs, T. Renger, *How Proteins Trigger Excitation Energy Transfer in the FMO Complex of Green Sulfur Bacteria*, Biophys. J. **91**, 2778 (2006).
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- [3] Y. Tanimura, Numerically “Exact” Approach to Open Quantum Dynamics: The Hierarchical Equations of Motion (HEOM), J. Chem. Phys. **153**, 020901 (2020).
- [4] A. Seneviratne, P. L. Walters, F. Wang, *Exact Non-Markovian Quantum Dynamics on the NISQ Device Using Kraus Operators*, ACS Omega **9**, 9666 (2024).
- [5] X. Dan, E. Geva, V. S. Batista, *Simulating Non-Markovian Quantum Dynamics on NISQ Computers Using the Hierarchical Equations of Motion*, J. Chem. Theory Comput. **21**, 1530 (2025).